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Individual document: Variability area

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Date:

May 21 - September 18, 2026

**Next-GenScientist: Blazar Variability and Time-Series
Analysis in Gamma-Ray Astronomy. Variability study
in a nutshell**

0.1 The research premise

“Mapping the Variability of Blazars: Building a FACT–HAWC Event Catalog” is an investigation project launched at “Next-gen Scientist” program looking for people with passion on topics like cosmology, astronomy, high energy physics, and data analysis. Blazars are a type of quasar, and they are active galactic nuclei which within its center have a supermassive black hole consuming the space matter surrounding it. It is in their absorption of matter, taken in relativistic speeds, where astrophysical mechanisms take place to exclude some of that matter in high energy jets of electromagnetic waves ranging from radio wavelength to the gamma ray spectra. The only difference between blazars and quasars, is that blazars are much easier to study because we are in the line of sight of the jet emissions, making us able to distinguish their change in ray flux. In a general sense, the spontaneous emission of these jets, referring to the timescale of their emission, the periodicity of these, and the magnitude of ray flux sparking from those sources, are commonly known as “flares”. The core idea of this investigation is to study the physical nature of these flares through mathematical models based on physical approaches. Understanding them, being able to characterize and cataloging them is a fundamental step unto grasping the true nature that lies within the kernel of these black holes and galactic centers. To accomplish this, the research relies heavily on the continuous data and observations provided by the HAWC and FACT observatories between the years 2009 and 2019, give or take. The two light sources studied are both the Markarian 501 and the Markarian 421 (mrk21 and mrk501 for short). Because of the earth’s rotation, and some other phenomena, both observatories have some time interval of light lectures, also known as transitions, for both of the sources; however, they have some key differences, and those are their time resolution. Since FACT and HAWC were built to attend different necessities, the former has a more clear time resolution (5-20 min), and it is sensitive to light emission, but the latter has a more standard 6 hour sidereal transits per day. Both have a cover of 6 hours per day, and gave us lectures on the TeV gamma ray wavelength. The question is, how to use this 12 hour daily data in order to understand more about the variability of light emissions and the subsequent flares so that we can better explain the complex physics happening in those blazars?

0.2 General Problem and Objectives

At the beginning of my participation, the project had gathered around 6 students from the physics major. The baseline to be capable of commencing with a more in depth participation or analysis for this project was:

- **Getting along with the investigation workflow and organization:** in specific, the group’s Notion page in which we would set the tasks, register any advancements, and establish the team’s overview of the current progress and workfronts of the various areas.
- **Proposing working divisions:** originally, it was set to be an investigation just along our research leader, Dr. José Andrés, and its Phd student Miguel. However, since the integration of 6 more people, it was imperative to divide the work into 3 main areas: data depuration and synthesis, variability, and literature and progress collectors. My

participation was more focused on the variability area, alongside my fellow peer and friend, Daniel Miranda.

- **Viewing the investigation progress:** hear from Dr. Andrés's, and Miguel's point of view the investigation's development and milestones; in addition to the current situation facing it.
- **Understanding the roles, and tools:** for the variability role, the whole point of it was to make some consensus about the definition of variability: how to conceptualize it and how to define it with a physical model. An extended search and study from previous papers was necessary. Computational methods and programs were also handled to the variability area for us to understand the idea of variability. Some basic knowledge and literature about cosmology, astrophysics and some other things were requested to begin with our involvement on the project.

0.3 Available and Learned Tools for the Research

Getting more into my personal experience on the variability area, and the research as a whole needs to be presented as a series of humble steps and basic comprehension of what an investigation and creation of a paper means. After having read some fundamental concepts and ideas of this area of physics, me and my peer were presented the gitlab repository in which Miguel has been working on since his inclusion in the project. Now in retrospect, his current undertaking was still really the most complicated and tedious struggle of most investigations: making sure the data was well enough curated, and manageable to begin with its prosecution. Since the FACT is an instrument that is exposed to the outside world, like a giant telescope, susceptible to weather conditions and much more, it was obligatory to make sure to correct the data in order to use it. This was more siding with the data analysis area of the research, nicknamed “prophet”, since it is the python library/tool used to curate the data.

With that being said, I will try now to talk about our “contributions” so that, whoever may read this, can have a sense of guidance, whatever that might mean to you. We were initially tasked to assimilate what progress was made on the variability studies for this investigation, and at the beginning we focused our efforts on what it would mean to work between me and Daniel in a programming environment. It is of really good practice to start knowing how to use git, github, and github desktop. These tools are the ground base for most engineering projects, so we commence our journey there (it does not have to be a formal course, youtube tutorials, and online guides will do the job). That, and we started with some basic tools like swagner-astro's “lightcurve” repository available online. After getting a feeling of those things, Dr. Andrés provided us with some papers on respect to the variability area called “Daily Monitoring of TeV Gamma-Ray Emission from Mrk 421, Mrk 501, and the Crab Nebula with HAWC” and the “Ornstein-Uhlenbeck parameter extraction from light curves of Fermi-LAT observed blazars”, while Miguel tasked us to review the previous two other methods to understand, and define variability. When someone asks about variability, the most common idea is how a variable fluctuates or has a spontaneous peak of some sort in an unexpected short time frame, that is what is the most important thing for the investigation,

since it is with that that we can study the rise time, decay time, asymmetry, and duration of a flare. In summary, the main two methods where:

1. **O-U method:** An approach using the stochastic Ornstein-Uhlenbeck method—a model describing the evolution of systems that exhibit mean reversion. The method appeared promising based on results obtained here and in another paper on pulsars. However, it was not pursued further after the model’s predictions showed mean-behavior distributions that differed strangely from what light curves typically demonstrate.
2. **STA/LTA method:** It is a technique based on seismology where you essentially have two, let’s say, averages running simultaneously: the short-term average and the long-term average. What exactly constitutes ”short-term” versus ”long-term”? That is determined experimentally, and you can define the window size as needed. The idea is that if the ratio between these two shows a sudden, sharp spike—either upward or downward—it could potentially indicate a flare (or perhaps a ”dip”?). However, you would have to adapt what short and long term mean in the context of the investigation, as well as some other things. They abandoned this STA-LTA seismology method (which is one we ”could” focus on) because the events weren’t clearly defined. They tested the method’s effectiveness for reporting events, but—given what Daniela (a collaborator) mentioned—perhaps not all of them were relevant, which is why they dropped it.
3. **Deutch collaborators:** In the investigation, Dr. José Andrés is working with some colleagues on Germany. The third approach was demonstrated using code written in German. Broadly speaking, the code performed a task similar to the one used for FACT: analyzing the light curve. Using data grouped by night—as well as data grouped into smaller time bins (such as 20 minutes, 2 minutes, half an hour, etc.)—the code matched the finer-grained data points to the corresponding days (e.g., identifying that a specific set of 20-minute data points belonged to a particular day, another set to the next, and so on). Each of these groups was then subjected to hypothesis testing to determine—based on a prediction or regression model—whether the light curve’s behavior could be characterized. If the behavior could be determined, the hypothesis was not rejected, implying an absence of significant variability.

After having studied, and understanding all the work done on the prophet side, we were left with reading a more recent paper called ”Characteristic Variability Timescales in the Gamma-Ray Power Spectra of Blazars”.

0.4 Areas for Improvement and Future Work

The most important thing that I could retrieve from all of my work can be told as the following: first, start to get along with the terminology and elemental concepts of astronomy, this field of physics, and the previous work that other investigations have done in order to present their definition of variability, since it depends on the research team to what will they call ”variability” and, thus, a flare; second, you can be doing this will performing the first one, try to analyse what progress has been made, and learn from the mistakes and setbacks that happened before you; and third, try to be calm about the whole thing, the idea is that

an investigation takes time, so one has to grasp most of what it is possible while working and committing errors of your own. Of what I am certain of, is that this has a lot to do with checking literature and other people's past papers. If it serves, a great way to define variability would be to have, as an assumption, that a time frame of the data behaves as a steady invariable light source with some standard deviation of flux. Given a statistical test (a comparison between the gamma ray flux being and not being constant, you calculate the difference between what the data provides and the supposition of constant flux) you should find that if there is a difference, it would be far greater than 0: there is variability in that time stamp. You can also look at Daniel's point of view for more information about the outlook of our "contributions".

In today's date, Miguel has forfeited the investigation as a whole, and is currently up to other students to continue with the project. Keep communication along the team, and address inquiries with genuine curiosity. The investigation attacks a really interesting and beautiful topic, but that also comes with the difficult and uncertain physics that lay hiding in plain sight from us.

1 Appendix: Key references and Mathematical Formulations used during this project:

References

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